Gotu AI Agriculture

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1 Tomato – planning and soil testing

1.1 Planning and Soil Testing for Tomato Cultivation

- **Topic Name and Purpose:** Crop planning and soil testing for tomato (*Solanum lycopersicum* L.) cultivation establish systematic protocols for site selection, soil fertility assessment, seasonal crop scheduling, and land preparation. These activities ensure optimal agro-climatic adaptation, effective disease prevention, balanced nutrient and water management, and maximized crop yield and fruit quality for both fresh market and industrial processing sectors.

- **Land and Site Selection:** Suitable land characteristics include medium black soils, loamy soils, or light to heavy well-drained soils with good topography. Select sunny fields with reliable water sources and effective surface drainage. Fields subject to prolonged waterlogging must be avoided because waterlogging causes root damage and promotes severe soil-borne diseases.

- **Climate and Environmental Requirements:** Tomato is a warm-season vegetable crop grown extensively in cool seasons as well. Extreme high or low temperatures cause poor seed germination, stunted vegetative growth, and poor fruit set. Warm and humid or subtropical climate conditions with optimal temperatures around 24–27 °C support robust plant development. Cloudy, cold, and drizzling weather increases susceptibility to foliar fungal pathogens such as late blight.

- **Soil Characteristics:** Ideal soil properties comprise medium black or loamy textures with a recommended pH range of 6.0–7.5, electrical conductivity (EC) below 2 dS/m, and organic carbon content exceeding 2.0% (with baseline soil health analysis indicating organic carbon above 0.75%). Fields must possess good natural drainage capacity.

- **Soil Sampling Procedures:** Representative soil samples must be collected across the target field prior to land preparation and planting to evaluate soil pH, organic matter, and baseline nutrient availability. Specific sampling depths, collection locations, sampling tools, and grid procedures are not specified in the provided sources.

- **Soil Sample Preparation and Submission:** Collected soil samples require proper handling, drying, mixing, labeling, and submission to recognized government, university, or NABL-approved soil testing laboratories accompanied by certified analyst documentation. Specific sample drying, packaging, and laboratory handling techniques are not specified in the provided sources.

- **Soil Testing Parameters:** Evaluated parameters include soil pH, electrical conductivity (EC), organic carbon percentage, primary macronutrients (Nitrogen, Phosphorus, Potassium), micronutrients (Zinc, Iron, Copper, Manganese), and Boron. Target soil nutrient removal benchmarks for a yield of 100 t/ha are 400 kg/ha Nitrogen, 110 kg/ha Phosphorus, and 550 kg/ha Potassium.

- **Soil Testing Methods:** Soil health analysis is conducted through standard laboratory testing procedures at accredited agricultural department or university facilities. Specific laboratory testing methods, chemical extraction reagents, testing equipment, and field testing kits are not specified in the provided sources.

- **Interpretation of Soil Test Results:** Documented target fertility ranges are pH 6.0–7.5, EC < 2 dS/m, organic carbon $> 2.0\%$ (minimum baseline $> 0.75\%$), available Phosphorus > 25 kg P/ha, available Potassium > 280 kg K/ha, Zinc > 0.6 ppm, Iron > 4.5 ppm, Copper > 0.2 ppm, and Manganese > 2.0 ppm. Recommended fertilizer doses must be modified according to specific soil test interpretations.

- **Fertilizer and Soil Amendment Recommendations:** Baseline nutrient requirements are moderately high at 100 kg/ha Nitrogen, 60 kg/ha P_2O_5, and 80 kg/ha K_2O. Apply well-decomposed farmyard manure (FYM) at 5.0 kg/m^2 along with neem cake at 200 g/m^2. Vermicompost addition at 1 kg/m^2 further enhances production. Biofertilizers include seedling root treatment with *Azospirillum* + PSB (20%) for 15 minutes at planting. Green manuring followed by *Trichoderma* application at 5 kg/ha within a week of ploughing is recommended. Nematode management options include neem cake and urea at 100 kg N/ha each at transplanting, or nursery carbofuran (0.4 g a.i./m^2) and field aldicarb (1 kg a.i./ha).

- **Land Preparation:** Prepare the field by deep ploughing (to a depth of 20 cm) and harrowing to break soil clods. Summer ploughing is recommended to promote pathogen desiccation. Incorporate FYM, neem cake, and organic compost into the soil. Construct raised beds with a bed width of 1.5 m and a 0.6 m pathway between beds, forming small planting pits on beds for enhanced root development.

- **Crop Planning:** Choose planting schedules based on district-specific conditions and production systems. In Maharashtra, sowing periods occur in May–June, September–October, and December–January, with transplanting in June–July (Kharif), October–November (Rabi/Rangda), and January–February (Summer). Crop duration ranges from 150 to 180 days. Seed rates are 400–500 g/ha for open fields (100 g/ha for specific hybrids), 350–400 g/ha under protected nurseries, and seedling densities of 13300–17700 plants per acre. Field spacing is 60–75 cm between plants and 75–90 cm between rows (or 1.0×0.8 m on raised beds). Select fresh market varieties (Dhanshree, Phule Kesari) or dual-purpose processing hybrids (Arka Apeksha, Arka Vishesh, Bhagyashree) based on disease resistance and market demand. Rotate crops with non-host crops like cereals, corn, sorghum, or millets.

- **Water and Irrigation Planning:** Source water from wells, tube wells, canals, or tanks, ensuring quarterly laboratory testing for pathogens. Drip irrigation is recommended to maximize water use efficiency and prevent direct foliage wetting. Daily water requirements under drip irrigation by stage are: sapling development

(1–2 mm/day, K_c = 0.2–0.3), plant development (2–3 mm/day, K_c = 0.4–0.6), flowering (3.5–4.0 mm/day, K_c = 0.7–0.8), fruit development (4.0–5.5 mm/day, K_c = 0.9–1.0), and fruit maturation (3.0–4.0 mm/day, K_c = 0.7–0.8). Calculate daily water requirement as WR = $E_{\text{pan}} \times K_c$.

- **Pre-planting Requirements:** Complete field preparation and clod crushing. Drench nursery raised beds with *Trichoderma* + *Pseudomonas* culture at 2% to manage damping-off. Treat 20–25 day old seedling roots by dipping for 15 minutes in a 10% *Trichoderma* + *Pseudomonas* culture (or 1% for 10 minutes) prior to transplanting.

- **Monitoring and Follow-up:** Reassess soil fertility through soil testing before applying recommended fertilizer doses. Conduct quarterly laboratory water testing for irrigation sources. Long-term soil fertility monitoring frequencies and reassessment protocols are not specified in the provided sources.

- **Effects on Crop Growth and Yield:** Proper soil condition, organic carbon, balanced NPK, and raised beds promote robust root development, plant vigor, fruit firmness (7.5–9.17 kg/cm^2), pericarp thickness (0.7–0.97 cm), and total soluble solids (4.5–5.3 °Brix). Productivity ranges from 40–60 t/ha for university varieties/hybrids and up to 83.0–93.2 t/ha for dual-purpose processing hybrids. Balanced drip irrigation prevents radial and concentric fruit cracking.

- **Limitations and Precautions:** Avoid fields with prolonged waterlogging due to damage to shallow root systems (restricted within 15–20 cm depth). Avoid dense seedbed sowing which causes weak, lanky seedlings. Do not apply fresh or slurry manure directly to crops; use only well-composted manure aged greater than one year. General agricultural guidance must not be assumed tomato-specific unless explicitly verified. Specific soil sampling errors are not specified in the provided sources.

- **Integrated Soil and Crop Planning:** Combine operations in a practical sequence: site selection on well-drained loamy/medium black soil; pre-season soil health testing; deep summer ploughing (20 cm) and green manuring with *Trichoderma* (5 kg/ha); incorporation of FYM (5.0 kg/m^2), neem cake (200 g/m^2), and vermicompost (1 kg/m^2); construction of 1.5 m raised beds with 0.6 m pathways; nursery seedling production (20–25 days) with bio-drenching (2%); seedling root dip in bio-culture (10%); transplanting at 60–75 cm × 75–90 cm spacing; drip irrigation installation with stage-wise Kc scheduling; and earthing-up at 30 and 45 days with organic or polythene mulching.

- **Other Relevant Information:** Light earthing-up is required at 30 days and 45 days after planting because tomato is a shallow-rooted crop with roots restricted to 15–20 cm soil depth. Mulching with straw or polythene conserves soil moisture, controls weeds, and enhances fruit quality and overall yield.